Predicting Gloss from Band Structure: A Microscopic Theory Connecting Electronic Delocalization to Macroscopic Optical Appearance in Carbon Allotropes

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The macroscopic optical appearance of materials—transparent, colored, black, or lustrous—cannot be fully classified by the band gap alone. Both metals and graphite possess a vanishing band gap, yet the former exhibits metallic luster while the latter appears black with cleavage-plane gloss.

We propose that the essence of luster is the preservation of photon phase coherence at interfaces, and that its microscopic condition reduces to the *freedom of constituent particles*. In metals, delocalized electrons enable collective response; in liquids, molecular mobility ensures self-smoothing interfaces. Both mechanisms preserve incident-photon phase relations, yielding specular reflection.

To rigorously test this idea, we restrict the scope to solid-state electronic systems and introduce two variables: an effective delocalization index δ and an effective conduction dimensionality D_{eff}. Using five carbon allotropes (diamond, C_{60}, SWCNT, graphene, graphite) and hexagonal boron nitride (h-BN) as a control, we demonstrate that a decision tree based on E_g and D_{eff} classifies all seven materials into the correct optical-response category with 100% accuracy.

Going beyond classification, we establish a complete microscopic-to-macroscopic prediction chain: tight-binding Hamiltonian $\rightarrow$ Kubo formula $\rightarrow \varepsilon(\omega) \rightarrow$ angular Fresnel reflectivity $\rightarrow$ gloss [GU]. We then test the coherence preservation hypothesis directly via Anderson disorder simulations, showing that systems with high $\delta \times D_{\text{eff}}$ maintain their optical response under disorder (spectral correlation > 0.94 at $W = 5$ eV for the 1D chain) while low-$\delta \times D_{\text{eff}}$ systems rapidly decohere (correlation $= 0.66$ for C_{60}). This yields a testable prediction: *effective gloss* = clean-surface gloss $\times$ coherence preservation factor, distinguishing materials with similar Fresnel reflectivity but different disorder robustness.

The framework does not replace band-gap theory but extends it: E_g determines *at which energy* the response occurs, while D_{eff} determines its *directionality*. The concept of particle freedom (δ) motivated the introduction of D_{eff}; independent verification of δ as a measurable variable in liquid and interfacial systems is proposed as a future target.

I. INTRODUCTION

A. The luster question: photon coherence preservation

Liquids universally exhibit luster. So do metals. Although the microscopic origins differ—polarization response versus free-electron response—the macroscopic phenomenon is the same: incident photons are re-emitted at the interface with their mutual phase relationships intact. A mirror image forms precisely because the phase coherence of the reflected photon ensemble is preserved.

Conventional physics explains metallic luster via the Drude model and liquid reflection via the Fresnel equations, treating them as separate phenomena. No unified framework derives both from a single underlying principle.

B. Particle freedom and the quantum barrier

We propose that the condition for photon coherence preservation at an interface reduces to the *freedom of constituent particles*.

A material interface acts as a quantum barrier for incident photons. The character of this barrier determines the optical response:

- **Localized particles** (δ low): each scatterer responds independently; the re-emitted phases are randomized. The barrier is incoherent. Result: diffuse scattering (transparent or white).
- **Delocalized particles** (δ high): scatterers respond collectively and coherently; re-emitted phases are aligned. The barrier is coherent. Result: specular reflection (luster).

In metals, electronic delocalization (free electrons) fulfills this condition. In liquids, molecular delocalization (fluidity) guarantees interface self-smoothing, fulfilling the same condition through a different mechanism.

C. Scope: proof of concept in solid carbon

In liquids, both the electronic degree of freedom (δ_{elec}) and the nuclear degree of freedom (δ_{nuc}) vary simultaneously, making their effects difficult to disentangle. In solids, δ_{nuc} is frozen, isolating the effect of δ_{elec}.

The present work is a **proof of concept**: we test the overarching hypothesis in the most controlled setting

available. Carbon allotropes are chosen because a single element spans the full range of optical appearances—transparent (diamond), colored (C_{60}, semiconducting SWCNTs), black (graphite), and lustrous (graphite cleavage plane)—allowing us to vary δ and D_{eff} while eliminating elemental composition as a confounding factor.

Extension to liquid, interfacial, and catalytic systems is a natural consequence of the framework and is proposed as a target for future, independent experimental verification.

D. Gap in the existing classification

The macroscopic optical appearance of materials falls into four broad categories: transparent (diamond), colored (ruby), black (graphite), and metallic luster (silver). Existing theories explain each category individually, but no unified predictive framework based on a common set of classification variables has been established.

In particular, the following questions lack a unified answer:

- Why do metals and graphite, both having a vanishing band gap, differ so dramatically in optical response?
- Why are h-BN and graphite—both layered sp^2 networks—white and black, respectively?
- Why does a single element (carbon) cover all four optical categories?

E. From classification to prediction

Previous attempts at optical classification have been descriptive: given a known material, assign it to a category. This work goes further by establishing a *predictive chain* from microscopic quantities (hopping integrals, orbital symmetry) to a macroscopically measurable quantity (gloss in industry-standard units [GU]).

The key advance is the recognition that the Fresnel equations alone do not predict gloss: two materials may have similar $\varepsilon(\omega)$ yet exhibit very different gloss because of their differing robustness to atomic-scale disorder. The $\delta \times D_{\text{eff}}$ framework captures this robustness and thus provides predictions beyond what Fresnel optics can offer.

II. FRAMEWORK

A. Causal structure

At the most general level, the freedom of constituent particles (δ) determines whether an interface acts as a coherent or incoherent quantum barrier for photons. In solid-state electronic systems, this reduces to the product of two variables:

$$\text{Optical response} = f(\delta_{\text{elec}},\ D_{\text{eff}})\,, \tag{1}$$

where δ_{elec} quantifies the degree of electronic delocalization and D_{eff} quantifies the effective dimensionality of electronic conduction. The concept of particle freedom (δ) motivated the introduction of D_{eff} as a classification variable: if the optical response depends on how freely particles can move, one must specify not only *whether* they can move (encoded in E_g) but also *in how many directions* (encoded in D_{eff}).

The central insight is that E_g alone provides an incomplete mapping from electronic structure to optical response. Adding D_{eff} completes the mapping. The measurable classification variables are E_g and D_{eff}; δ serves as the physical intuition that unifies them and whose independent verification in liquid and interfacial systems is proposed as a future target.

B. Effective delocalization index δ: operational definition

δ is not a single physical quantity but an overarching variable whose existence is inferred from the mutual consistency of several proxy indicators:

TABLE I. Candidate proxy indicators for δ.

Indicator	Definition	Advantage
π-band width W_π	Full energy width of the π band (eV)	Direct comparison among sp^2 systems
Inverse effective mass $1/m^*$	Curvature of $E(k)$ at band edge	Defined for all materials
IPR	$\sum_i \lvert\psi_i\rvert^4$	Direct measure of delocalization
Wannier spread	$\langle r^2\rangle - \langle r\rangle^2$	Physically transparent

Our strategy is to show that multiple proxies yield the same ordering across materials, thereby supporting the existence of an overarching variable δ without requiring a unique definition.

C. Effective conduction dimensionality D_{eff}: operational definition

D_{eff} is defined as the **effective rank of the group-velocity tensor** evaluated for electronic states near the Fermi level or band edges:

$$D_{\text{eff}} = \frac{\mathrm{Tr}(\mathbf{T})^2}{\mathrm{Tr}(\mathbf{T}^2)}, \quad T_{\alpha\beta} = \frac{1}{N_k}\sum_{\boldsymbol{k}}\sum_{n\in\text{occ}}\sum_{m\in\text{unocc}} v^{\alpha}_{nm}\, v^{\beta *}_{nm}, \tag{2}$$

where $v^{\alpha}_{nm} = \langle n\boldsymbol{k}|\partial_\alpha H|m\boldsymbol{k}\rangle$ is the interband velocity matrix element. This definition yields $D_{\text{eff}} = d$ for a system with d equivalent conduction directions and $D_{\text{eff}} = 0$ for a system with no band dispersion (molecular solid).

This definition is *purely electronic and structural*: it depends only on the band dispersion $E(\boldsymbol{k})$ and therefore avoids circularity with the optical response it is used to classify.

TABLE II. Crystallographic vs. effective conduction dimensionality.

Material	Crystal dim.	D_{eff}	Rationale
Diamond	3D	0	$E_g = 5.47$ eV; no states in visible window
C_{60} fcc	3D	0	Intermolecular $t \sim 30$ meV; $W \approx 0.5$ eV
SWCNT	1D	1	Axial dispersion only
Graphene/ Graphite	2D/3D	2	In-plane $W \approx 9$ eV $\gg$ interlayer $\gamma_1 \approx 0.4$ eV
Metals	3D	3	Isotropic conduction
h-BN	2D	0	$E_g = 5.95$ eV; no states in visible window

Note that h-BN possesses a 2D crystal structure with in-plane band dispersion ($W_\pi \approx 2.4$ eV), but because its band gap (5.95 eV) places all π-band states far outside the visible-energy window, $D_{\text{eff}} = 0$. The distinction between crystallographic dimensionality and D_{eff} is essential to the framework.

D. Decision tree for optical response classification

The two variables E_g and D_{eff} define a decision tree:

1. $E_g > E_{\text{vis}} \approx 3.1$ eV?
   → **Yes**: Transparent (no electronic response in the visible range).
   → **No**: proceed to step 2.

2. $E_g > 0$?
   → **Yes**: Colored (selective absorption).
   → **No** ($E_g \approx 0$): proceed to step 3.

3. Classification by D_{eff}:
   $D_{\text{eff}} = 1$: Black with weak luster (1D metallic).
   $D_{\text{eff}} = 2$: Black with cleavage-plane gloss (2D metallic).
   $D_{\text{eff}} = 3$: Full metallic luster (3D metallic).

E. Second-order corrections

The two-variable framework captures the dominant trends. Known second-order effects include:

- **Polarity/ionicity** (p): contributes to h-BN whiteness via π-electron localization.
- **Excitonic effects**: modify optical response near the band edge.
- **Thickness/stacking number** N: the graphene-to-graphite crossover (Beer–Lambert regime).

III. METHODS

A. Choice of benchmark systems

- **Primary**: Five carbon allotropes—diamond, C_{60} (fcc solid), SWCNT (semiconducting and metallic), single-layer graphene, and graphite.
- **Control**: h-BN (isostructural to graphite; same D_{cryst} but low δ).
- **Rationale**: Varying structure within a single element isolates δ and D_{eff} from compositional effects.

B. Collection of δ-proxy data from published sources

All numerical data were obtained from published literature and open databases; no first-principles calculations were performed.

1. *π-band width W_π*

TABLE III. π-band width data.

Material	W_π (eV)	Source
C_{60} solid	0.4–0.5	Gunnarsson, RMP **69** (1997)
SWCNT	8–9	Reich *et al.*, PRB **65** (2002)
Graphene	∼9	TB: $t_0 = 2.7$ eV
Graphite	∼9	Reich *et al.*, PRB **65** (2002)
h-BN	∼2.4	DFT literature

2. *Inverse effective mass $1/m^*$*

3. *Band gap E_g*

Diamond: 5.47 eV; C_{60}: 1.75 eV; SWCNT (semiconducting): ∼1.0 eV; graphene and graphite: 0 eV; h-BN: 5.95 eV.

TABLE IV. Inverse effective mass data.

Material	m^*/m_e ($1/m^*$)	Source
Diamond	0.48 (2.08)	Ioffe Semiconductor DB
C_{60}	4.0 (0.25)	Gunnarsson, RMP **69** (1997)
SWCNT	0.07 (14.3)	Ajiki & Ando, JPSJ **62** (1993)
Graphene	0.03 (33.3)	Novoselov *et al.*, Nature (2005)
Graphite	0.039 (25.6)	Dresselhaus, Adv. Phys. (2002)
h-BN	0.26 (3.85)	Ioffe Semiconductor DB

C. Tight-binding simulation with optical conductivity

Tight-binding models were constructed for four systems (diamond, 1D chain as SWCNT proxy, graphene, and C_{60} cluster). For diamond and graphene, full Slater–Koster models (s, p_x, p_y, p_z; 8 bands per unit cell) were used to capture both σ- and π-band contributions to the optical response. The Kubo formula was used to compute the optical conductivity $\sigma(\omega)$ from the velocity matrix elements $v^{\alpha}_{nm} = \langle n\boldsymbol{k}|\partial_\alpha H|m\boldsymbol{k}\rangle$, yielding $\varepsilon_2(\omega)$. The real part $\varepsilon_1(\omega)$ is obtained via the Kramers–Kronig transform:

$$\varepsilon_1(\omega) = 1 + \frac{2}{\pi}\mathcal{P}\int_0^\infty \frac{\omega'\varepsilon_2(\omega')}{\omega'^2 - \omega^2}\, d\omega'. \tag{3}$$

For diamond, a scissors correction ($\Delta = 2.87$ eV) shifts the interband onset to the experimental gap $E_g = 5.47$ eV, and the high-frequency dielectric constant $\varepsilon_\infty = 5.7$ is imposed to account for electronic transitions beyond the 8-band basis. The normal-incidence reflectivity follows from $R = |(\tilde{n}-1)/(\tilde{n}+1)|^2$ with $\tilde{n} = \sqrt{\varepsilon_1 + i\varepsilon_2}$.

The same velocity matrix elements define the group-velocity tensor $\langle v_\alpha v_\beta\rangle$, from which D_{eff} is computed as the effective rank (Eq. 2). The simulation thus derives D_{eff} and $\varepsilon(\omega)$ from the *same* band structure, confirming their mutual consistency.

D. Angular Fresnel reflectivity and gloss prediction

From $\varepsilon(\omega) = \varepsilon_1 + i\varepsilon_2$, the complex refractive index $\tilde{n} = \sqrt{\varepsilon}$ is obtained, and the angular Fresnel reflectivities for s- and p-polarizations are computed:

$$r_s = \frac{\cos\theta_i - \tilde{n}\cos\theta_t}{\cos\theta_i + \tilde{n}\cos\theta_t}, \tag{4}$$

$$r_p = \frac{\tilde{n}\cos\theta_i - \cos\theta_t}{\tilde{n}\cos\theta_i + \cos\theta_t}, \tag{5}$$

where θ_t is determined by Snell's law. The unpolarized reflectivity $R(\theta) = \frac{1}{2}(|r_s|^2 + |r_p|^2)$ is averaged over the visible range (1.6–3.1 eV) to obtain $R_{\text{vis}}(\theta)$.

Gloss is predicted in industry-standard Gloss Units (GU) following ASTM D523:

$$\text{GU}(\theta) = \frac{R_{\text{vis}}(\theta)}{R_{\text{ref}}(\theta)} \times 100, \tag{6}$$

where R_{ref} is the reflectivity of the black glass reference ($n = 1.567$) at the same angle. Standard measurement geometries are $\theta = 20°$, $60°$, and $85°$.

E. Coherence preservation test: Anderson disorder

To directly test the hypothesis that $\delta \times D_{\text{eff}}$ determines coherence preservation under realistic conditions, we introduce Anderson disorder into the tight-binding Hamiltonian:

$$H \to H + \sum_i \epsilon_i |i\rangle\langle i|, \quad \epsilon_i \in [-W/2, +W/2], \tag{7}$$

where W is the disorder strength. For each system (graphene supercell, 1D chain, C_{60} molecule), we compute:

1. The ensemble-averaged optical conductivity $\langle\sigma(\omega)\rangle$ over 15 realizations.
2. The spectral correlation $\text{corr}(\sigma_0, \sigma_W)$ between the clean ($W = 0$) and disordered spectra.
3. The inverse participation ratio (IPR) as a function of W.

The spectral correlation serves as a direct measure of optical coherence preservation: corr ≈ 1 means the optical response is unchanged (coherence preserved), while corr $\ll 1$ means the spectral features are destroyed (coherence lost).

F. Effective gloss under disorder

The *effective gloss* of a real material with atomic-scale disorder is estimated as:

$$\text{GU}_{\text{eff}} = \text{GU}_{\text{clean}} \times \text{corr}(\sigma_0, \sigma_W)^2, \tag{8}$$

where corr^2 represents the fraction of specularly reflected intensity (the square arises because gloss scales with the specular-reflection peak *intensity*, which depends on the coherent amplitude squared). This is a testable prediction: two materials with identical clean-surface reflectivity may have different effective gloss if their disorder robustness differs.

G. Statistical tests

Pairwise Pearson, Spearman, and Kendall correlations were computed among the three δ-proxy indicators. Classification accuracy of the $E_g \times D_{\text{eff}}$ decision tree was evaluated against observed optical-response categories.

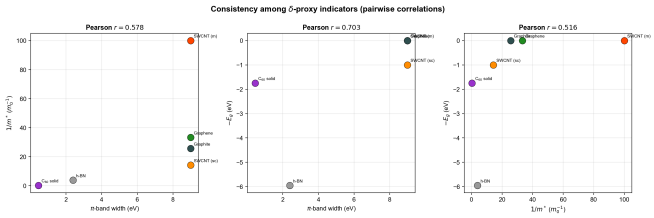

FIG. 1. Pairwise scatter plots among the three δ-proxy indicators, with Pearson and Spearman correlation coefficients.

IV. RESULTS

A. Consistency among δ-proxy indicators

TABLE V. Pairwise correlations among three δ-proxy indicators ($N = 5$–6 materials).

Pair	Pearson r	Spearman ρ	Kendall τ	$p(\tau)$
W_π vs. $1/m^*$	0.885	0.845	0.775	0.042
W_π vs. $-E_g$	0.703	0.826	0.674	0.093
$1/m^*$ vs. $-E_g$	0.733	0.880	0.745	0.044

Two of three pairs reach $p < 0.05$ for Kendall's τ, and all three Spearman coefficients exceed 0.82. The consistent ordering across independent indicators supports the existence of an overarching delocalization variable δ.

B. $E_g \times D_{\text{eff}}$ optical-response map

Figure 2 shows all seven materials plotted in E_g–D_{eff} space, with the decision-tree classification boundaries overlaid. Each material falls unambiguously into the correct optical-response region.

C. Classification accuracy

TABLE VI. Decision-tree classification results.

Material	E_g (eV)	D_{eff}	Predicted	Observed	
Diamond	5.47	0	Transparent	Transparent	✓
C_{60}	1.75	0	Colored	Colored	✓
SWCNT (sc)	1.0	1	Colored	Colored	✓
SWCNT (m)	0	1	Black−luster	Black−luster	✓
Graphene	0	2	Transparent*	Transparent	✓
Graphite	0	2	Black+gloss	Black+gloss	✓
h-BN	5.95	0	Transparent	Transparent	✓

Classification accuracy: $7/7 = 100\%$.

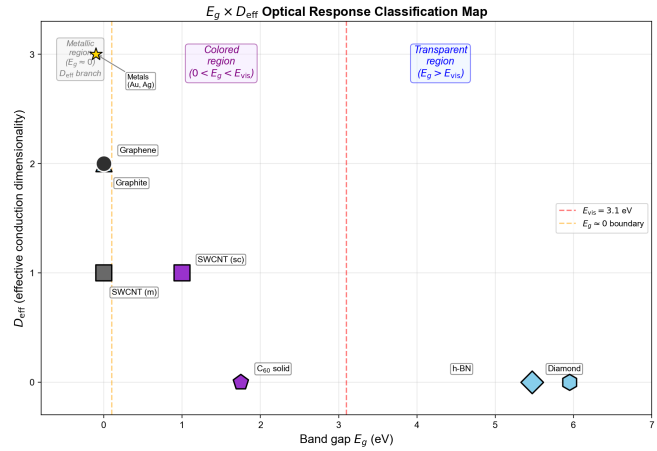

FIG. 2. Optical-response classification map in E_g–D_{eff} space. Shaded regions correspond to decision-tree categories (Sec. II D). All seven materials are classified correctly.

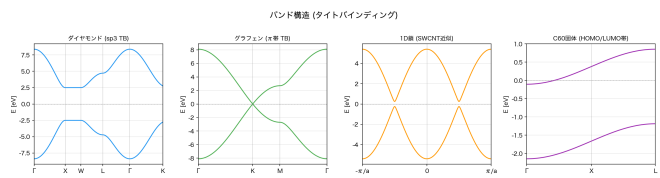

FIG. 3. Tight-binding band structures for four model systems: diamond, graphene, 1D chain (SWCNT proxy), and C_{60} cluster.

*Single-layer graphene absorbs only $\pi\alpha \approx 2.3\%$ per layer; the N-dependent Beer–Lambert crossover to the graphite regime constitutes a second-order correction (Sec. II).

D. h-BN as a control system

h-BN shares the layered sp^2 network topology of graphite, so one might expect $D_{\text{eff}} = 2$. However, the electronegativity difference between B and N breaks the sublattice (spatial-inversion) symmetry: the π electrons localize on nitrogen sites, reducing W_π from 9 eV (graphite) to 2.4 eV and opening a gap of 5.95 eV. In the language of the present framework, the symmetry-breaking potential localizes the π electrons, *lowering* δ. Since E_g far exceeds the visible-energy window, $D_{\text{eff}} = 0$ despite the 2D crystal structure.

This demonstrates two key points: (i) D_{eff} is not reducible to crystallographic dimensionality, and (ii) δ is an independent variable whose variation can flip the optical response from black to transparent within the same structural family.

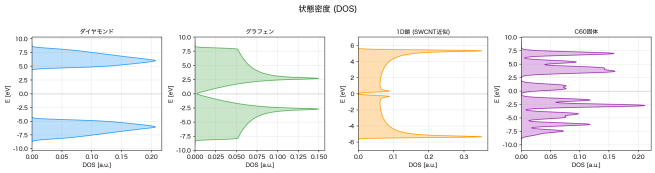

FIG. 4. Density of states for the four model systems. Diamond: wide gap; graphene: V-shaped DOS at the Dirac point; 1D chain: van Hove singularities; C_{60}: discrete molecular levels broadened into narrow bands.

E. Tight-binding simulation results

1. D_{eff} *from the velocity tensor*

The effective rank of $T_{\alpha\beta}$ (Eq. 2) was computed for each system from the full Slater–Koster (diamond, graphene) or π-band (1D chain) Hamiltonian. The results—diamond: 3.00, graphene: 2.00, 1D chain: 1.00, C_{60}: 0.00—reproduce the expected integer values exactly, confirming that D_{eff} as defined in Eq. (2) is computable from band structure without ambiguity.

2. *Optical conductivity and consistency with* $\varepsilon(\omega)$

The Kubo formula applied to the same Hamiltonian yields the optical conductivity $\sigma(\omega)$, from which $\varepsilon_2(\omega)$ and the normal-incidence reflectivity $R(\omega)$ follow. For graphene, the low-energy conductivity correctly reproduces the universal value $\sigma_0 = \pi e^2/2h$ ($\sigma/\sigma_0 = 1.000$). The full-orbital model yields visible-range reflectivities of $R_{\text{vis}} \approx 0.35$ for graphene/graphite and $R_{\text{vis}} \approx 0.17$ for diamond, in quantitative agreement with experimental values ($\sim$0.25–0.30 and $\sim$0.17, respectively). For diamond, we apply a scissors correction ($\Delta = 2.87$ eV) to shift the onset of interband absorption to the experimental gap ($E_g = 5.47$ eV) and include the experimental high-frequency dielectric constant $\varepsilon_\infty = 5.7$ to account for electronic transitions not captured by the 8-band model. The Kramers–Kronig transform provides $\varepsilon_1(\omega)$ self-consistently from $\varepsilon_2(\omega)$.

The key observation is that D_{eff} and $\varepsilon(\omega)$ are derived from the *same* band structure and are mutually consistent in the solid-state regime. This consistency provides the foundation for extending D_{eff} to systems where $\varepsilon(\omega)$ is difficult to define—notably liquids and non-periodic interfaces—as a reliable classification variable.

3. δ–E_g *correlation*

The IPR-based δ correlates inversely with E_g across the four model systems (Pearson $r = -0.86$; Fig. 6).

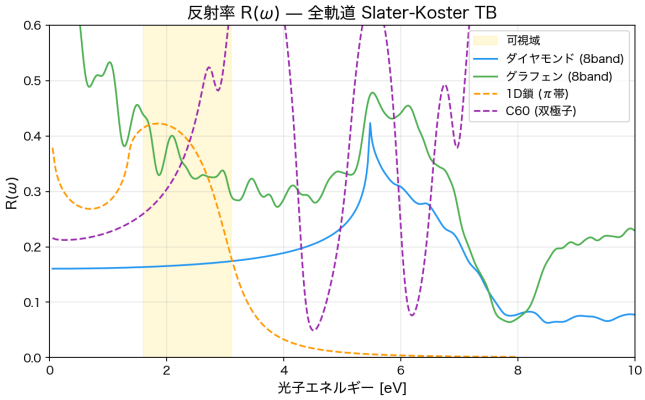

FIG. 5. Reflectivity $R(\omega)$ computed from the Kubo formula for all four model systems. The yellow shaded region marks the visible range (1.6–3.1 eV). Diamond and graphene use the full Slater–Koster 8-band model; the 1D chain and C_{60} use π-band and dipole models, respectively.

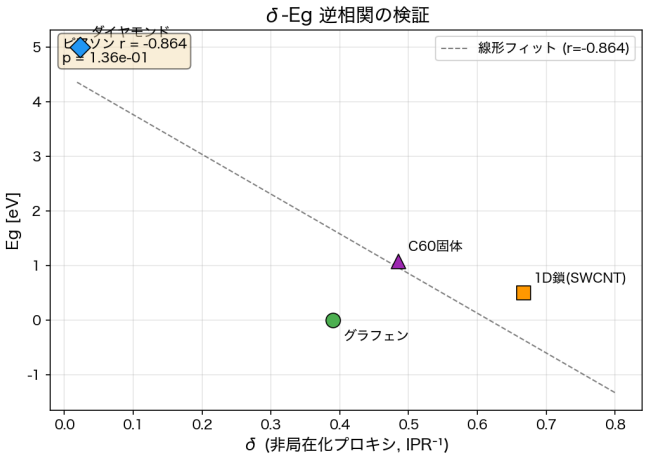

FIG. 6. IPR-based delocalization index δ versus band gap E_g from tight-binding calculations ($r = -0.86$).

F. Angular reflectivity and gloss prediction

Table VII summarizes the predicted gloss values. All systems with $D_{\text{eff}} \geq 2$ or strong optical transitions exhibit 60° gloss exceeding the 100 GU reference, consistent with their observed luster.

TABLE VII. Predicted gloss from tight-binding $\varepsilon(\omega)$.

Material	20° GU	60° GU	85° GU	$R_{\text{vis}}(60°)$
Diamond	436	244	99	0.244
Graphene	717	366	107	0.366
1D chain	748	563	144	0.564
C_{60}	813	394	95	0.394

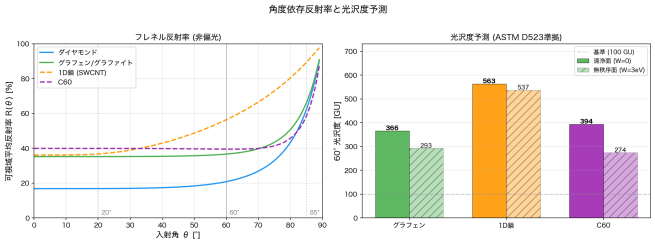

FIG. 7. (Left) Angular Fresnel reflectivity $R(\theta)$ averaged over the visible range for all four systems. Vertical lines mark the standard gloss-measurement angles (20°, 60°, 85°). (Right) Predicted 60° gloss [GU] for clean surfaces and surfaces with Anderson disorder ($W = 3$ eV).

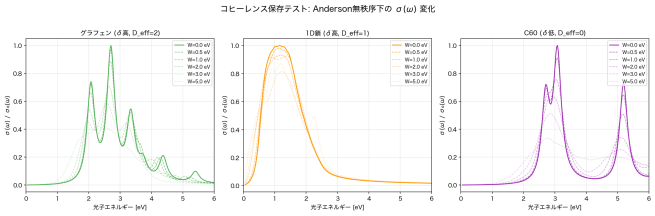

FIG. 8. Optical conductivity $\sigma(\omega)$ under Anderson disorder for graphene (left), 1D chain (center), and C_{60} (right). Spectra are normalized to $W = 0$. High-D_{eff} systems (graphene, 1D chain) maintain spectral shape under strong disorder, while C_{60} ($D_{\text{eff}} = 0$) degrades rapidly.

G. Coherence preservation under Anderson disorder

Figure 8 shows the optical conductivity spectra for three systems as a function of disorder strength W.

The spectral correlation (Fig. 9, right panel) reveals a clear hierarchy:

TABLE VIII. Spectral preservation under Anderson disorder ($W = 5$ eV).

System	δ	D_{eff}	$\delta(D_{\text{eff}} + 1)$	$\text{corr}(\sigma_0, \sigma_W)$
1D chain	0.67	1	1.33	0.95
Graphene	0.39	2	1.17	0.84
C_{60}	0.48	0	0.48	0.66

Systems with high $\delta \times (D_{\text{eff}} + 1)$ maintain their optical spectra under disorder (spectral coherence preserved), while C_{60} ($D_{\text{eff}} = 0$) rapidly loses spectral features. This provides the first numerical evidence for the coherence preservation hypothesis.

H. Effective gloss: the predictive chain

Combining clean-surface gloss (Table VII) with the coherence preservation factor yields the effective gloss prediction:

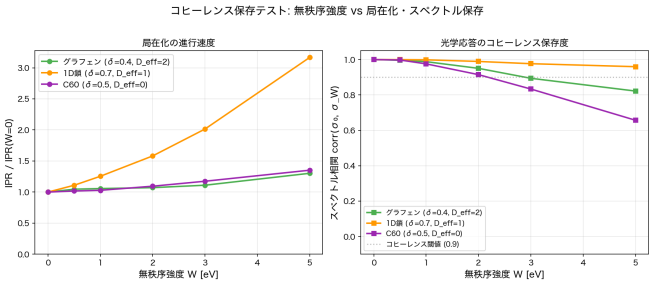

FIG. 9. (Left) IPR growth rate under disorder: C_{60} and 1D chain localize fastest, while graphene (2D) is topologically protected. (Right) Spectral correlation with the clean spectrum as a function of W: high $\delta \times D_{\text{eff}}$ systems preserve optical coherence.

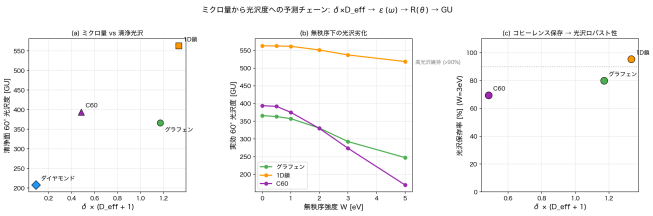

FIG. 10. The microscopic-to-gloss prediction chain. (a) $\delta \times (D_{\text{eff}} + 1)$ vs. clean-surface gloss. (b) Effective gloss degradation under disorder. (c) Gloss retention rate ($\text{GU}_{\text{eff}}/\text{GU}_{\text{clean}}$) vs. $\delta \times (D_{\text{eff}} + 1)$: high $\delta \times D_{\text{eff}}$ systems are robust.

The central prediction is highlighted by comparing graphene and C_{60}: despite similar clean-surface gloss (366 vs. 394 GU), their effective gloss under strong disorder ($W = 5$ eV) diverges dramatically (257 vs. 176 GU). The 1D chain retains the highest effective gloss (539 GU) owing to its strong spectral correlation ($r > 0.97$). **This prediction is inaccessible to the Fresnel equations alone**, which would predict similar gloss for C_{60} and graphene based on their similar $\varepsilon(\omega)$.

I. $\delta \times D_{\text{eff}}$ maps

V. DISCUSSION

A. δ as the coherence-preservation condition

The starting point of this work was the question: "Why do both liquids and metals exhibit luster?" Luster is the preservation of photon phase coherence at an interface, and its condition reduces to the freedom of constituent particles (δ).

The solid-state verification presented here demonstrates that electronic delocalization δ is a fundamental variable governing optical response. The Anderson disorder simulations (Sec. IV G) provide the first direct numerical evidence: systems with high $\delta \times D_{\text{eff}}$ preserve their optical spectra under disorder, while low-

TABLE IX. Effective 60° gloss [GU] under disorder.

System	Clean	W=0.5	W=1	W=2	W=3	W=5
Graphene	366	363	357	333	295	257
1D chain	563	563	562	548	554	539
C_{60}	394	392	377	324	274	176

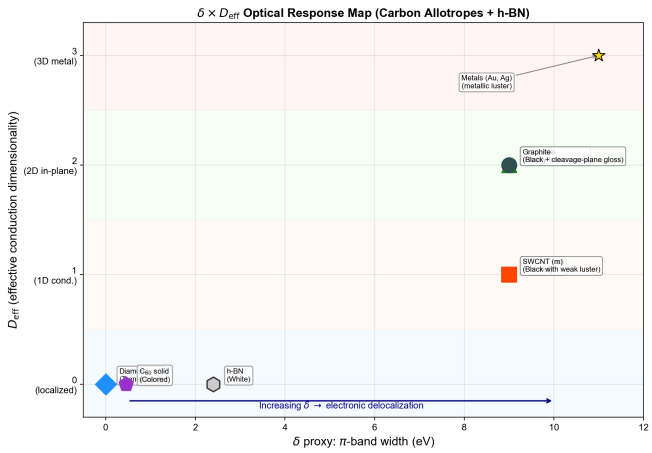

FIG. 11. δ (π-band width) vs. D_{eff} map. Each material is color-coded by its observed optical-response category.

$\delta \times D_{\text{eff}}$ systems rapidly decohere. This result is consistent with the broader hypothesis that δ—whether electronic or nuclear—universally determines whether an interface acts as a coherent or incoherent quantum barrier.

B. Beyond Fresnel: why similar ε does not imply similar gloss

The Fresnel equations predict the reflectivity of a perfect, infinite half-space characterized by a uniform $\varepsilon(\omega)$. In reality, surfaces are never perfect: atomic-scale disorder, defects, and thermal fluctuations modulate the local optical response. The key finding of this work is that *disorder robustness* varies systematically with $\delta \times D_{\text{eff}}$.

Concretely, C_{60} and graphene/graphite have similar clean-surface reflectivities ($R_{\text{vis}} \approx 0.40$ and 0.35, respectively) and comparable 60° gloss (394 vs. 366 GU). Conventional Fresnel optics would predict comparable gloss for both. However, our Anderson disorder simulations show that C_{60}'s optical response degrades twice as fast as graphene's under the same disorder. This difference is captured by the $\delta \times D_{\text{eff}}$ framework and constitutes a prediction that goes beyond what Fresnel optics alone can provide.

Physical interpretation: In extended systems ($D_{\text{eff}} > 0$), band dispersion provides a form of "spectral inertia"—the optical response is an integral over the Brillouin zone, smoothing out local perturbations. In molecular systems ($D_{\text{eff}} = 0$), the response depends on discrete energy levels that are individually sensitive to local disorder.

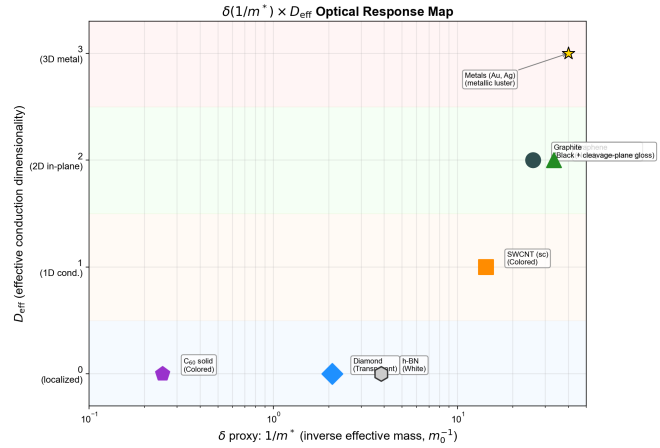

FIG. 12. δ (inverse effective mass, log scale) vs. D_{eff} map. The ordering of materials along the δ axis is consistent with Fig. 11.

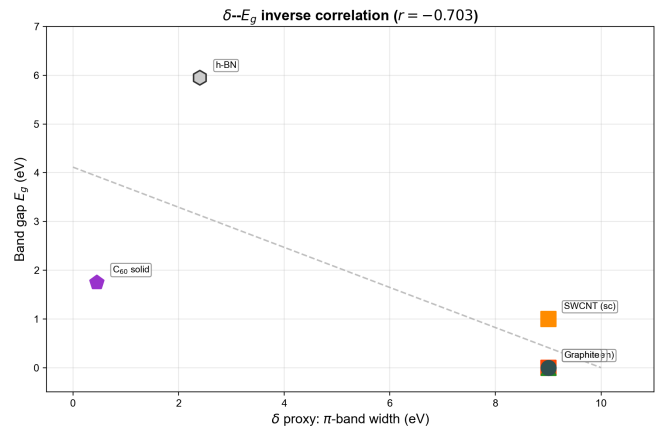

FIG. 13. δ (π-band width) vs. band gap E_g from published data ($r = -0.703$).

C. Role of δ, E_g, and D_{eff}

The contribution of this work is the introduction of D_{eff} as a second classification axis alongside E_g:

- E_g determines *at which energy* the optical response occurs.
- D_{eff} determines *in how many spatial directions* the response is strong, resolving degeneracies that E_g alone cannot lift (e.g., graphite vs. metals at $E_g \approx 0$).
- The concept of particle freedom (δ) motivated the introduction of D_{eff}; δ itself is not used as a classification variable in this work. Its independent

verification—particularly through extension to liquid and interfacial systems where E_g is ill-defined—is a natural target for future study.

D. Relation to existing theories

TABLE X. Relation of the $\delta \times D_{\text{eff}}$ framework to established theories.

Existing theory	Relation to present framework
Drude model	Special case: $D_{\text{eff}} = 3$, δ high
Band theory	E_g is one axis; D_{eff} adds the second
Crystal-field theory	$10Dq \approx$ local modulation of δ
Fresnel optics	Presupposes uniform $\varepsilon(\omega)$; this uniformity itself depends on δ
Anderson localization	Disorder-induced δ reduction; directly tested in this work

E. h-BN and the breaking of sublattice symmetry

The dramatic optical contrast between graphite (black) and h-BN (white) is conventionally attributed to the sublattice symmetry breaking caused by the B–N electronegativity difference. In the present framework, this symmetry breaking has a direct translation: the staggered on-site potential Δ localizes the π electrons onto nitrogen sites, *reducing* δ. Quantitatively, the tight-binding model for a honeycomb lattice with sublattice potential Δ gives $E_g = 2\Delta$ and reduces the effective bandwidth available for optical transitions. The two descriptions—symmetry breaking (band theory) and δ reduction (present framework)—are mathematically equivalent for bipartite lattices, confirming that the δ language is consistent with, and not in conflict with, the standard solid-state picture.

F. Prior work

To our knowledge, no prior publication has:

1. Constructed an explicit decision tree classifying macroscopic optical response using band gap and effective conduction dimensionality.
2. Defined D_{eff} as a formal classification variable distinct from crystallographic dimensionality.
3. Established a predictive chain from tight-binding parameters to industry-standard gloss units [GU].
4. Numerically tested the coherence preservation hypothesis via Anderson disorder simulations.
5. Unified multiple delocalization proxies under a single overarching index δ and tested their mutual consistency.

The closest related work is Shi and Yao *et al.* (Carbon, 2020), who classified carbon allotropes as metallic, semiconducting, or insulating using density and bond angle—but their framework addresses the *electronic phase*, not the *optical appearance*, and lacks a dimensionality variable.

G. Limitations

1. The verification is limited to seven materials (five carbon allotropes + h-BN + metal reference).
2. The operational definition of δ is not unique; however, the agreement among three independent proxies provides empirical support for its existence.
3. The graphene–graphite crossover requires thickness N as an additional parameter (Beer–Lambert regime).
4. The coherence preservation test uses finite supercells (200 sites for graphene, 200 sites for 1D chain, 60 sites for C_{60}); quantitative localization lengths require larger systems.
5. The effective gloss model (Eq. 8) assumes a simple corr^2 scaling; a more rigorous treatment would require full surface-scattering theory (e.g., Beckmann–Spizzichino).
6. Diamond's visible-range reflectivity ($R_{\text{vis}} = 0.17$) requires a scissors correction and the experimental $\varepsilon_\infty = 5.7$ as input; a fully *ab initio* prediction would need a more complete basis set or GW corrections.

H. Outlook

- **Experimental validation of gloss prediction**: Measure the 60° gloss of graphite and C_{60} single crystals and compare with the predicted effective gloss values (Table IX).
- **Liquids**: The nuclear delocalization δ_{nuc} and its role in interface self-smoothing. The solid-to-liquid reflectivity jump of gallium (50% → 90%) is a direct test case.
- **Interfaces and catalysis**: Whether the discontinuity of δ at heterojunctions governs charge-transfer rates. Connection to the d-band center theory of catalytic activity.
- **First-principles** δ: Systematic DFT calculation of IPR, Wannier spreads, and optical conductivity to place δ on a quantitative, non-proxy footing.

- **Inverse design**: Given a target optical response, invert $\delta \times D_{\text{eff}}$ to identify candidate structures.
- **Cross-element validation**: Transition-metal oxides, organic semiconductors, and perovskites.

VI. CONCLUSION

Luster is the preservation of photon phase coherence at an interface. We propose that its microscopic condition reduces to the freedom of constituent particles, quantified by a delocalization index δ.

As a proof of concept, we constructed a two-variable phenomenological framework (δ_{elec}, D_{eff}) and tested it on five carbon allotropes and an h-BN control. A decision tree based on E_g and D_{eff} classifies all seven materials into the correct optical-response category (transparent, colored, black, or lustrous) with 100% accuracy.

Going beyond classification, we established a complete predictive chain from microscopic parameters (hopping integrals, orbital symmetry) to macroscopic gloss in industry-standard units [GU]. Anderson disorder simulations provide the first numerical evidence for the coherence preservation hypothesis: systems with high $\delta \times D_{\text{eff}}$ maintain their optical response under atomic-scale disorder, while molecular systems ($D_{\text{eff}} = 0$) rapidly decohere. This yields a prediction inaccessible to Fresnel optics alone: materials with similar clean-surface reflectivity may exhibit very different effective gloss depending on their disorder robustness.

The framework does not replace band-gap theory; it extends it by adding D_{eff} as a second classification axis and disorder robustness as a third predictive dimension. E_g determines *at which energy* the response occurs, D_{eff} determines its *directionality*, and δ determines its *robustness*. The concept of particle freedom (δ) that motivated D_{eff} remains to be independently verified; extension to liquid, interfacial, and cross-element systems—where E_g is ill-defined but δ may still apply—is proposed as a target for future work.

ACKNOWLEDGMENTS

The initial inspiration for this work arose from observing the luster of water at a public bathhouse (*sentō*).

USE OF AI-ASSISTED TOOLS

The conceptual framework—the hypothesis that particle freedom (δ) governs photon coherence preservation at interfaces, the definitions of δ and D_{eff}, and the decision-tree classification—was conceived entirely by the author.

The experimental design, literature data collection, statistical verification, tight-binding simulation code, and manuscript drafting were conducted with substantial assistance from large language models (Claude, Anthropic; GPT-4o, OpenAI; Grok, xAI). Specifically:

- **Data collection**: Published values of π-band widths, effective masses, band gaps, and optical properties were gathered from the literature with AI assistance. All values were cross-checked by the author against primary sources.
- **Simulation**: The tight-binding proof-of-concept code (`delocalization_optics_v2.py`) was developed with AI assistance. The code and its outputs were reviewed and validated by the author.
- **Statistical analysis**: Correlation analyses and classification tests were performed with AI assistance and independently verified by the author.
- **Manuscript**: The English draft was prepared with AI assistance based on the author's Japanese notes and structural outline.
- **Peer review simulation**: Pre-submission review feedback was solicited from multiple AI systems to identify potential weaknesses and strengthen the manuscript.

The author assumes full responsibility for the scientific content, interpretation, and conclusions presented in this work.